

App Dev Project Report

1. Student Details

Name: Rahul Pathak

Roll Number: 23f2000798

Email: 23f2000798@ds.study.iitm.ac.in

About Me:

I am a Data Science and Business student at IIT Madras with a strong interest in AI and machine learning. I enjoy building real-world projects that combine business and technology.

2. Project Details

Project Title: Placement Portal Application

Problem Statement:

To design and build a web-based Placement Portal that allows Admin (Institute), Companies, and Students to interact efficiently. The system should handle company approvals, placement drives, student applications, and tracking of placement status while avoiding manual processes like spreadsheets and emails.

Approach:

The project was developed using Flask as the backend framework with a structured and modular approach. First, the database schema was designed using SQLAlchemy, defining relationships between User, Company, Student, Placement Drive, and Application tables.

Next, authentication and role-based access control were implemented using Flask-Login. Then, separate dashboards were created for Admin, Company, and Student roles. Core features like company approval, drive creation, application submission, and status tracking were built step by step.

Finally, the frontend was developed using Jinja2 templates and styled using Bootstrap and custom CSS to ensure a clean and responsive user interface.

3. AI/LLM Declaration

I used ChatGPT (GPT-5) to assist in debugging, improving code structure, and understanding Flask concepts such as routing, database relationships, and authentication flow.

The extent of AI/LLM usage is approximately 20–25%, mainly for guidance and optimization.

All final implementation, logic building, debugging, and integration were done manually.

4. Technologies and Frameworks Used

Technology / Library	Purpose
----------------------	---------

Flask	Backend web framework
SQLAlchemy	ORM for database handling
SQLite	Database
Jinja2	Template engine
Bootstrap	UI styling
CSS	UI enhancement
Flask-Login	Authentication and session management
WTForms	Form validation
Werkzeug	Password hashing

5. Database Schema / ER Diagram

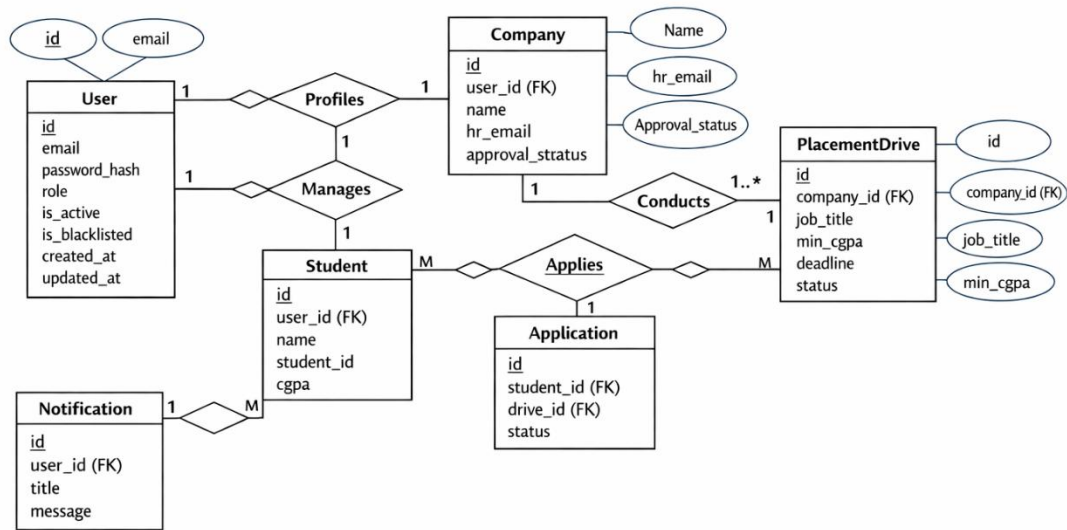
Tables:

- **User** → stores login credentials and role (admin/company/student)
- **Company** → stores company details and approval status
- **Student** → stores student profile and academic details
- **PlacementDrive** → stores job/drive information
- **Application** → stores student applications
- **Notification (optional)** → stores user notifications

Relationships:

- One-to-One → User → Company
- One-to-One → User → Student
- One-to-Many → Company → PlacementDrive
- One-to-Many → Student → Application
- One-to-Many → PlacementDrive → Application

Additionally, a **unique constraint** ensures that one student cannot apply to the same drive multiple times, as implemented in the database model.



6. API Resource Endpoints

(No external APIs implemented)

All functionalities are handled through Flask routes (server-side rendering using Jinja templates).

Examples of routes:

Route	Method	Description
/login	GET, POST	User login
/register/student	GET, POST	Student registration
/register/company	GET, POST	Company registration
/drives	GET	View placement drives
/apply/	GET	Apply for drive
/admin/approve-company	GET	Approve company
/company/create-drive	GET, POST	Create drive

7. Architecture and Features

Architecture Overview:

- app.py → Main Flask application and routes
- models.py → Database schema and relationships
- forms.py → Form validation
- templates/ → HTML pages using Jinja2
- static/ → CSS and UI assets

Core Features Implemented:

- Role-based authentication (Admin, Company, Student)
- Company approval system by Admin
- Placement drive creation and approval
- Student application system
- Application status tracking (Applied, Shortlisted, Selected, Rejected)
- Placement history tracking
- Search functionality for admin
- Blacklisting system for users

Additional Features:

- Duplicate application prevention using database constraint
- Clean and responsive UI
- Flash messages for user feedback
- Secure password hashing
- Profile editing for students and companies

8. Video Presentation

Drive Link: https://drive.google.com/file/d/1NYhnRDyd9Y2hBYMORRqvUAkZ_CmM-fXC/view?usp=sharing